Benjamin Scully

Benjamin Scully | b.scully [at] mail.utoronto.ca | 0009-0005-0313-3129

Education

University of Toronto *2025 - Current*

PhD - Astronomy and Astrophysics

Thesis topic: Understanding neutron star flashes through observation and simulation
Supervisors: Professors Marten H. van Kerekwijk & Bart Ripperda

University of British Columbia *2020 - 2025*

BSc Combined Honours in Physics and Astronomy | with Distinction; Co-operative Education Program

Thesis title: Investigating the Detectability of AQN Dark Matter Emission in Modern Telescopes
Supervisor: Professor Ludovic van Waerbeke

Publications

1. Alarcon, R., et al. incl. **Scully, B.**, The DarkLight Experiment at TRIUMF. *Submitted to NIM A*, arXiv: 2609.05674 (Sep. 2026)
2. Sekatchev, M., Liang, X., Majidi, F., **Scully, B.**, Waerbeke, L. V., & Zhitnitsky, A., The Glow of Axion Quark Nugget Dark Matter: (III) the Mysteries of the Milky Way UV Background. *JCAP*: 10.1088/1475-7516/2026/03/010, arXiv: 2504.15382 (Mar. 2026)
3. Soni, S., et al. incl. **Scully, B.**, LIGO Detector Characterization in the first half of the fourth Observing run. *CQG*: 10.1088/1361-6382/adc4b6 arXiv: 2409.02831 (Sep. 2024)
4. **Scully, B.** , Matzner, C. D., & Yalinewich, A., Observability of flashes from ejecta crashes in aspherical supernovae, with application to SN 2008D. *MNRAS*: 10.1093/mnras/stad2360 arXiv: 2307.15859 (Aug. 2023).

Research Experience

Graduate Research Assistant - *UofT* *May 2026 - Present*

Supervised by Dr. Bart Ripperda

- Employing resistive relativistic magnetohydrodynamic simulation to study magnetic reconnection around neutron stars.

Graduate Research Assistant - *UofT* *Oct 2025 - Present*

Supervised by Dr. Marten Van Kerkwijk

- Using interstellar scintillometry to analyze Green Bank Telescope observations of the Crab pulsar to reveal mechanism for poorly understood emissions.

Undergraduate Honours Thesis - *UBC* *Sep 2024 - Jul 2025*

Supervised by Dr. Ludovic Van Waerbeke & Dr. Ariel Zhitnitsky

- Modeling Axion Quark Nugget dark matter model emission in JWST and Euclid bands to test observation potential. Found that the signal is theoretically detectable using component separation.

Undergraduate Research Assistant - *TRIUMF* *May - Aug 2024*

Supervised by Dr. Jason Holt

- Implemented a new quantum operator in the group-wide code base and identified other necessary improvements in the field of quark mixing.

Undergraduate Research Assistant - *TRIUMF* *Sep 2023 - May 2024*

Supervised by Dr. Katherine Pachal

- Improved mechanical design and assembled scintillators, detector architecture, and wiring of fast scintillator detectors for the DarkLight experiment. Achieved sub-200 picosecond time resolution.

Undergraduate Research Assistant - *LIGO UBC* *May - Aug 2023*

Supervised by Dr. Jess McIver

- Studied signal vs glitch classification in LIGO detectors to improve detection capabilities. Identified gaps in classifier using machine learning techniques in Python with *KERAS & TensorFlow*

Undergraduate Research Assistant - *University of Toronto* *May - Aug 2022*

Supervised by Dr. Christopher D. Matzner

- Studied observability of circumstellar ejecta collisions from aspherical supernovae and compared my simulations to observed events. Achieved novel conclusions on observables and published in MNRAS.

Awards and Grants

Canada Graduate Research Scholarship - Master's – (*$27 000 over 1 year*) *2026-2027*
Faculty of Arts & Science Top Doctoral Award – (*$180 000 over 4 years*) *2025-Present*
Paul Sykes Scholarship in Astronomy – (*$1 200*) *2024*
Dante Ciccone Memorial Scholarship in Astronomy – (*$1 575*) *2024*
Dean of Science Scholarship – (*$280*) *2024*
Canadian Institute for Nuclear Physics URS – (*$11 500*) *2024*
NSERC USRA – (*$11 000*) *2023*
Charles and Jane Banks Scholarship – (*$270*) *2022*
NSERC USRA – (*$9 600*) *2022*
Dean's Scholar – UBC *2021-2025*

- Awarded to students with an average of $\geq 90\%$ the previous Winter Session.

Presentations and Talks

1. **Undergraduate Thesis** *Apr 2025*
   Axion Quark Nugget Glow: Observing dark matter in modern telescopes
2. **CUPC 2024 Undergraduate Talk** *Oct 2024*
   Standard Model Mixology: Exploring Quark Mixing Through Nuclear Theory
3. **TRIUMF Science Week "Science Pitch" (Honourable Mention)** *Jul 2024*
   Up is Down: Ab-Initio Approaches to Superallowed Beta Decay to Test for New Physics
4. **UBC Summer-student Astronomy Colloquium** *Aug 2023*
   Semi-supervised clustering of LIGO-Virgo signals to improve GSpyNetTree classification

Posters

1. **TRIUMF Co-op Poster Presentations** *Aug 2024*
   Standard Model Mixology: Exploring Quark Mixing Through Nuclear Theory
2. **TRIUMF Co-op Poster Presentations** *Apr 2024*
   DarkLight Fast Scintillator Trigger Detector Development
3. **UofT Dunlap SURP Poster Presentation (Honourable Mention)** *Aug 2022*
   Do circumstellar collisions make observable transients in aspherical supernovae?

Outreach and Service

Toronto Public Library (TPL) Outreach Facilitator *Feb 2025 - Present*

- Delivering astronomy workshops and other programs through TPL branches to public of all ages

Astrotours Logistics Coordinator (UofT Astronomy Graduates) *Oct 2025 - Present*

- Supporting co-directors in planning astrotours events. Stand in for any absent astrotours chairs.

Mediation Committee (UofT Astronomy Graduates) *Oct 2025 - Present*

- Acting to mediate disputes within the department. Advocates for a fair graduate stipend annually.

UBC PHAS EDI Committee *May 2024 - May 2025*

- Member of the UBC Physics & Astronomy Department's Equity Diversity and Inclusion committee.
- Responsible for establishing professional development and effective for members of the community.

Girl Guide Physics Activities *Jan 2024 - Apr 2025*

- Taught interactive physics activities to Girl Guide groups of ages 5-12.
- Activities included LED circuits, spaghetti towers, and instruction about general relativity .

UBC Canadian Undergraduate Physics Conference Organizer *Jan - Oct 2024*

- Acted as external coordinator in the organization of CUPC at UBC for the Fall of 2024.
- Handled all aspects of securement, and organization of all nineteen keynote and panellist speakers.
- Managed catering and sponsorship details, and judged student presentations.

UBC Physics Olympics *Mar 2023 - Mar 2025*

- Formulated rules, designed and ran pre-build competition in 2024 and 2025.
- Ran vacuum pump pre-build competition and Quizzics (Physics Quiz) competition in 2023.

UBC Faraday show *Nov 2023*

- Presented to over 300 members of the public to introduce physics to younger generations.

Teaching Experience

Stars and Stars and Planets (AST221) Teaching Assistant – UofT *Sep - Dec 2026*

- Conducting office hours, tutorials and exam review sessions. Grading assignments and exams.

Stars and Galaxies (AST201) Teaching Assistant – UofT *Jan - May 2026*

- Conducted office hours and exam review sessions. Reviewed exam questions for accuracy.

The Sun and its Neighbours (AST101) Teaching Assistant – UofT *Sep - Dec 2025*

- Ran office hours, held telescope observation sessions, and conducted exam review sessions.

Skills

Programming: Python, C++, ROOT, Java, Fortran, Matlab

Applications: GitHub, Linux terminal, LaTeX, MadGraph, SolidWorks, Microsoft Office

Other: Object oriented design, Data collection & analysis, Computing clusters, Scientific writing, Delicate materials handling, Physical construction/assembly

Attributes: Detail oriented, Communication, Public speaking, Teamwork, Empathy

Workshops and Special Programs

Canadian Astroparticle Physics Summer School (CAPSS) *May 2023*

French Immersion Dual Dogwood high school diploma *Jun 2020*

Languages

Fluent: English, French – **Learning:** Spanish (B1)